Hareendra Sri Nag Nerusu

+1 (352) 246-7446 | hareendra05@gmail.com | linkedin/Hareendra | **GitHub** | LeetCode

Education

University of Florida, Florida, United States **August 2024 - May 2026**

Master of Science, Computer Science ***GPA: 3.92 / 4.00***

- Courses**:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Distributed Systems, Database Systems, Software Engineering, Cloud Computing

Skills

Languages & Scripting: Java, Python, Go, C++, JavaScript, TypeScript, SQL, Bash, Shell Scripting

Web & Frameworks: HTML5, CSS3, Bootstrap, React.js, Redux, React Router, JSON, XML, Axios, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Resilience4j

Backend & Databases: RESTful APIs, Microservices, Spring Core, Spring Web, Spring Cloud, Spring Cloud Gateway, Apache Kafka, JDBC, Oracle, SQL Server, MySQL, PostgreSQL, MongoDB

Cloud, DevOps & Monitoring: AWS (EC2, Lambda, S3, RDS, IAM, CloudWatch), Docker, Kubernetes, Terraform, Jenkins, CI/CD, Maven, Gradle, Docker Compose, Prometheus, Grafana, ELK Stack, SLF4J, Logback

Testing & Tools: JUnit, Mockito, React Testing Library, Postman, Git, GitHub, GitLab, Jira, Confluence, Windows, Linux, macOS

Experience

Software Development Engineer **Jan 2025 - Present**

University of Florida *Gainesville, Florida*

- **Delivered** a high-performance medical web application by designing asynchronous **FastAPI microservices** that scaled to support **3K+ concurrent users**, achieving a **40%** increase in request throughput through optimized API contract design.
- Built and optimized distributed data storage using **AWS S3** for large objects and **PostgreSQL on AWS RDS** for transactional metadata, applying indexing and schema tuning to **reduce query latency by 30%** while maintaining data consistency and fault tolerance.
- Implemented secure, versioned **REST APIs** with **OAuth2 and JWT** and introduced **Redis caching** for frequently accessed data, **cutting database reads by 45%** and **improving end-to-end API response times by 25%** under load.
- Owned the full service lifecycle by containerizing applications with **Docker**, automating **CI/CD using GitHub Actions**, deploying on **AWS EC2 and Lambda**, and enabling **CloudWatch-based monitoring and logging**, which **reduced deployment failures by 25%** and improved production observability and on-call readiness.

Software Development Engineer **May 2022 - Dec 2023**

C Square Solutions *Bangalore, India*

- Engineered backend **microservices in Java 17 using Spring Boot**, exposing REST APIs consumed by multiple client applications and reliably serving **10K+ daily requests** across **5+ production systems**.
- Accelerated data access by refactoring **PostgreSQL and MySQL** queries with **Spring Data JPA and JDBC**, introducing indexing and optimized join strategies that **cut API response times by 35%** in high-traffic services.
- **Insisted on the highest standards** for service security by architecting a standardized authentication layer using **Spring Security** and **JWT**, which eliminated recurring production vulnerabilities and **reduced defect rates by 20%** across multiple production systems.
- Streamlined release workflows by implementing **Maven- and Jenkins-based CI/CD pipelines**, containerizing services with **Docker**, and deploying on **AWS (EC2, RDS)**, **improving deployment consistency and reliability** across environments.

Projects

GatoRides – Ride-Sharing Web Application

- Engineered stateless ride-matching and authentication services using **Golang REST APIs** and **MongoDB**, sustaining **100+ concurrent requests** during load testing and **improving booking success rate by 35%** through efficient request handling and concurrency control.
- **Lowered end-to-end latency by 30%** by optimizing **MongoDB indexing strategies**, reducing API payload sizes, and integrating backend services with a **React frontend** for real-time ride status updates.

Task Management Dashboard & Notification Platform (Event-Driven)

- Built a customer-facing **React dashboard** integrated with **FastAPI microservices**, enabling task creation, filtering, and lifecycle tracking, **reducing manual task coordination effort by 30%** for active users.
- **Implemented event-driven task notifications** using **AWS SQS and Redis-backed workers**, decoupling synchronous APIs from background processing and **improving API responsiveness by 25%** under peak load.

Medical Imaging Prediction Platform (ML-backed System)

- **Simplified** complex research workflows by engineering an automated ML inference pipeline using **PyTorch** and **FastAPI**, which reduced experiment turnaround time by **40%** through the strategic reuse of trained artifacts and automated execution.
- Engineered end-to-end **ML data pipelines** using **NumPy** with versioned dataset and model storage in **AWS S3**, enabling consistent evaluation across runs and **preventing invalid or non-reproducible model outputs**.